

ARAVIND KRISHNAN

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PROFILE

Motivated by a passion for innovation and technology, combining strategic thinking with consistent effort to deliver impactful results. Confident enough in programming languages like Python and Java, with growing expertise in AI technologies including Machine Learning and Deep Learning. Comfortable using tools like Anaconda Navigator, and recognized for strong teamwork, problem-solving, and critical thinking abilities.

EDUCATION

BTech CSE & AI , AMRITA VISHWA VIDYAPEETHAM,AMRITAPURI CGPA: 6.49 2022-Present

12th Standard , RMHSS MELATTUR May 2022

SKILLS

Technical Skills	Python
Soft Skills	Teamwork, Critical Thinking
Cloud	Google Cloud Platform
Libraries and Frameworks	NumPy , Matplotlib , HTML
Tools and Technologies	VS Code, Anaconda Navigator

PROJECTS

Disaster Response Robot *December, 2024*

- Collaborated in a team of 4 & Engineered a prototype of remotely operated robotic system tailored for high-risk environments.
- Integrated ROS-based teleoperation for seamless navigation and precise control, enabling safe exploration in hazardous zones.
- Learned the importance of safe, remote-controlled exploration in hazardous environments.

Generative Adversarial Networks for Handwritten Digit Generation *December, 2024*

- Developed a Generative Adversarial Network (GAN) utilizing Transpose CNNs to generate high-resolution images from random noise.
- Designed and implemented a generator network with learned upsampling using transpose convolutions for realistic image synthesis.
- Trained the model using Binary Cross-Entropy (BCE) loss and the Adam optimizer, achieving visually coherent and high-quality outputs.
- Applied Batch Normalization to stabilize training and improve convergence.
- Evaluated model performance using FID Score and inception score, optimizing the architecture for better image realism.
- Understood implications of adversarial techniques in security, including their role in deepfakes and data manipulation. This experience sparked further interest in adversarial machine learning and its impact on cybersecurity.

Smart Water Quality Monitoring System *July, 2023*

- Developed an IoT-based Smart Water Quality Monitoring System to assess and report water quality in real-time, ensuring safe and healthy water standards.

- Implemented sensors for pH, turbidity, and temperature to monitor water parameters.
- Designed a dashboard for data visualization, providing users with instant feedback on water quality.
- Utilized NodeMCU for seamless data transmission to a cloud server, allowing real-time data access and remote monitoring.
- Ensured system reliability through regular calibration and error-checking protocols.
- Familiarized with dashboard-based visualization and cloud connectivity for remote monitoring systems.

AI-Based Indoor Navigation System for Visually Impaired

Ongoing

- Developing a hardware-software prototype using Raspberry Pi to provide real-time voice guidance and Haptic for the visually impaired.
- Implemented Real-Time Scene Understanding using computer vision models to detect obstacles and safe paths in dynamic indoor environments.
- Focusing on Safety-Critical AI, ensuring low-latency processing and high reliability for the user physical safety.

CERTIFICATES

DataCamp Courses on Deep Learning in Python

- Designed and trained deep learning models
- Developed deep learning workflows from data preprocessing to model evaluation

COINCENT

- Successfully completed a comprehensive program that enhanced my skills in cybersecurity practices and ethical hacking techniques.
- Duration: 01st Jul, 2023 – 15th Aug, 2023

Understanding Machine Learning

- Gained a solid understanding of machine learning, including key concepts .
- Supervised and unsupervised learning, model evaluation, deep learning, and applications in computer vision and natural language processing.

COMMUNITY OUTREACH PROGRAMS

SSR Project on Indian Culture and Heritage

October, 2024

Conducted an interactive session on Indian Culture and Heritage for about 150 students at Amrita Vidyalayam, Pandalam, as part of a Student Social Responsibility (SSR) initiative, fostering cultural awareness and appreciation. managing session flow, participant engagement, and overall coordination and Explained about Mahabharatham.

VOLUNTEERING ACTIVITIES

Saptati of Amma, Sri Mata Amritanandamayi Devi

Selfless and unwavering commitment, contributing to an event that impacted thousands, embodying the true spirit of service to humanity.

Vidyut'25 Volunteer

My duty involves managing and finding Sponsors for the Vidyut'25 multinational fest. This event helped me increase my communications skills along with my ability to manage tasks efficiently.

Black Hat Asia 2025

Though I come from an AI/ML background, going to Black Hat helped me see how skills like anomaly detection, data privacy, or secure communication are relevant.